

# Threat-hunting methodology

Authored by Argyro Ouzounoglou. Public portfolio edition.

The two example IP addresses have been replaced with documentation addresses.

This document demonstrates a hunting workflow, not a measured business outcome.

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# CYBER THREAT HUNTING & THREAT INTELLIGENCE OPERATIONS

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*BUILDING PROACTIVE DEFENSE THROUGH THREAT HUNTING, CTI, AND ELASTIC SIEM INTEGRATION*

# STRATEGIC IMPLEMENTATION & OPERATING GUIDE

**T**his document establishes the baseline for our Threat Hunting and Cyber Threat Intelligence (CTI) program. It defines:

- ▶ What threat hunting and CTI are
- ▶ How we will operate (methodology, roles, outputs)
- ▶ How CTI feeds hunting and how hunting feeds detection engineering
- ▶ How Elastic SIEM and Jira will be used to standardize execution and reporting

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## 1. EXECUTIVE SUMMARY

Threat Hunting is the **proactive** practice of searching for adversaries in the environment using available telemetry (endpoints, authentication, network, etc.). It is not driven by alerts alone; it is driven by hypotheses and investigative analysis, often under an **Assume Breach** mindset.

### Why It Is Critical:

- **Dwell Time Reduction:** Advanced persistent threats (APTs) can remain undetected for days or weeks (averaging 10–16 days). Hunting identifies these threats in early stages, reducing the window for data exfiltration.
- **Security Gap Identification:** It uncovers misconfigurations, unapproved software, and missing logging infrastructure that automated tools miss.
- **Defense-in-Depth:** It serves as a necessary layer of protection when signature-based or behavioral rules in AV/EDR fail to trigger.

### Expected Outcomes:

Improved SOC maturity, reduced organizational risk, faster incident response, and the transformation of "hunts" into permanent automated detections.

## 2. CURRENT STATE & GAP ANALYSIS

For a low-medium maturity SOC, several key gaps must be addressed to transition **from** reactive **to** **proactive** operations:

- **Mindset Gap:** Hunting is proactive and hypothesis-led, not random querying or alert chasing.
- **Knowledge Gap:** Effective hunting focuses on adversary behavior (TTPs) and how to observe it in logs.
- **Data Gap:** Baseline "normal" behavior is undefined, making it impossible to identify "abnormal" deviations.
- **Integration Gap:** Cyber Threat Intelligence (CTI) is often viewed as a list of "bad IPs" (low-value IOCs) rather than actionable tactical intelligence. CTI needs to be treated as **purpose + context + prioritization** that drives hypotheses and detection.

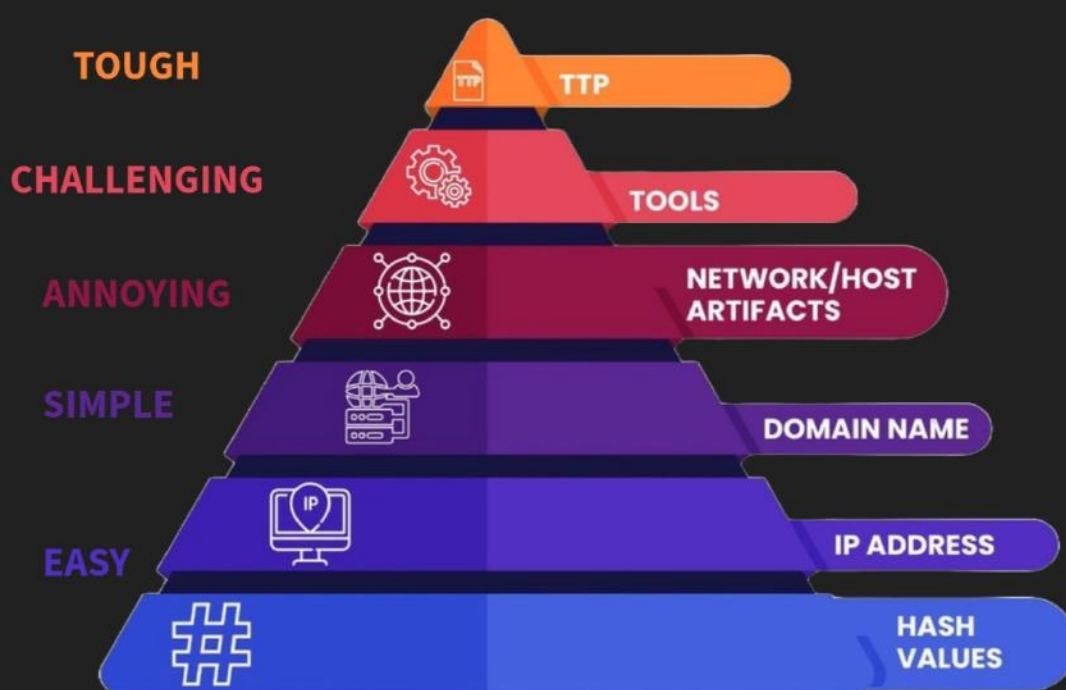
## 3. THREAT HUNTING FUNDAMENTALS (STANDARD TERMINOLOGY)

### 3.1 TTPs & ATT&CK (What we hunt)

- **TTPs (Tactics, Techniques, and Procedures):**
  - a. **Tactics:** The adversary's goal (e.g., Lateral Movement).
  - b. **Techniques:** How they achieve the goal (e.g., PSEXEC).
  - c. **Procedures:** The specific sequence of actions. describe how adversaries operate (tactics, techniques, procedures).
- **MITRE ATT&CK** is used to map behaviors and build hunts/detections, it is not "just IOC lookup."

### 3.2 Pyramid of Pain (What we prioritize)

A model showing that while file hashes (easy to change) are low-value indicators, **TTPs and tools** are the "toughest" for an attacker to change, making them the most effective targets for hunting.



### 4. THREAT HUNTING TYPES

- **Hypothesis-Driven (Structured):** Starts with a specific TTP (e.g., "An attacker is using DLL search order hijacking").
- **Intel-Driven (Unstructured):** Triggered by specific Indicators of Compromise (IOCs) like hashes or IPs provided by CTI.
- **Data-Driven (Situational/Analytics):** Analyzing logs for anomalies without a specific threat in mind (e.g., "Does anything in our network look suspicious?").

## 5. BUILDING THE BASELINE (ELASTIC-FOCUSED)

In the Elastic Stack, we define "normal" to isolate the "signal" from the "noise."

### Elastic Dashboards & KPIs:

- **Visualizations:** Use Elastic "Lens" to track volume spikes in process.args length or unique destination IPs per workstation.
- **KPIs:** Number of hunts transitioned to detections, average time to identify anomalies, and percentage of environment coverage.

Per the hunting strategy guidance, baseline must explicitly document and maintain visibility into:

- **Running processes**
- **User logins** (where/who/when/type)
- **Network connections**
- **Services and scheduled tasks**
- **Permitted/approved software**

This baseline is the reference point for identifying outliers and abnormal patterns.

| Focus Area     | What "Normal" Looks Like                                                          |
|----------------|-----------------------------------------------------------------------------------|
| Authentication | Users logging in from known geolocation during business hours.                    |
| Network        | DNS traffic on Port 53. Web traffic on 443/80.                                    |
| Endpoint       | Known parent-child process relationships (e.g., services.exe spacing svchost.exe) |

## 6. THREAT INTELLIGENCE INTEGRATION (CTI)

### 6.1 Threat Intelligence Source Selection

Selecting and governing intelligence sources is **critical** to ensure **relevance**, **reduce noise**, and **prevent operational overload** while enabling accurate, intelligence-driven hunting and detection.

#### Intelligence is collected from:

- Open-source intelligence (OSINT)
- Commercial intelligence feeds
- ISAC / Government sources
- Internal telemetry (primary validation source)

#### All intelligence must:

- Normalize + enrich before correlation (GeoIP, WHOIS, tags)
- Use standards (STIX/TAXII) when applicable
- Verify external IOCs with internal telemetry
- Avoid “correlation overload” (don’t link everything)

#### Operational pipeline:

**Collection** → **TIP** → **SIEM** → **Enrichment** → **Detection/Hunt Trigger**

## 6.2 Intelligence Levels

We will move beyond simple IOC matching to tactical intelligence integration.

### CTI Levels:

- **Strategic:** High-level trends (e.g., "Ransomware targeting our sector") for leadership.
- **Tactical:** TTPs used by specific actors (e.g., "APT28 uses cdn-assets-update[.]com") for hunters.
- **Technical:** Specific hashes and IPs for automated SIEM blocking.

### Operationalizing in Elastic:

- **Enrichment:** Use Elastic's "Enrich" processor to tag incoming logs with MITRE ATT&CK tags from CTI reports.
- **Hunting Trigger:** Use a CTI report about "Lazarus Group using PSEXec" to trigger a hypothesis-driven hunt in the SIEM.

## 6.3 CTI Prioritization (Relevance, Actionability, Criticality)

We cannot act on all intel. We must define prioritization using:

- **Relevance** (aligned to our assets/sector/tech/geography)
- **Actionability** (can be used to detect/block/investigate)
- **Criticality** (risk if unmitigated, impact factors)

## 6.4 Correlation methods (How CTI becomes actionable)

| Correlation Techniques       | Examples                                                                                                                                                                                   |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exact matching               | IP 192.0.2.10 have been seen in both the organization's SIEM and in the subscribed threat intel feed.                                                                                      |
| Infrastructure pivoting      | The domain name "evil.login[.]com" -> host on 198.51.100.20 -> hosts 3 other known malicious domains.                                                                                      |
| Time-Based Correlation       | Based on event timing to reconstruct incidents                                                                                                                                             |
| Fuzzy matching               | "dropbox[.]com" vs "dropb0x-secure[.]com", malware hashes with 93% similarity                                                                                                              |
| TTP and Campaign Correlation | <ul style="list-style-type: none"><li>• Powershell command -&gt; T1059.001</li><li>• Infrastructure overlaps with APT28</li><li>• Confirms behavioral linkage to known adversary</li></ul> |

## 7. THREAT HUNTING METHODOLOGY

### The Hunting Lifecycle:

1. **Hypothesis Creation:** "Adversaries are abusing PowerShell to download payloads".
2. **Data Collection:** Pull Winlogbeat and Sysmon data (Event IDs 4104 for PowerShell ScriptBlock logging).
3. **Analysis:** Use Elastic KQL to search for EncodedCommand or Invoke-WebRequest.
4. **Detection Creation:** If malicious activity is found, create a permanent Elastic Detection Rule.
5. **Feedback Loop:** Refine the rule to reduce false positives based on SOC feedback.

## 8. DETECTION ENGINEERING

The CTI methodology explicitly states ATT&CK is used for:

- Threat Hunting
- Detection Engineering
- Threat emulation/red teaming

Every hunt must end in one of:

- Detection candidate (validated behavior → rule)
- Logging gap (visibility deficiency)
- Baseline update (normal behavior documented)

## 8.1 Detection Engineering Elastic + Sigma

The goal of every hunt is a permanent detection.

- **Sigma Rules:** We will write generic Sigma rules (e.g., for "Suspicious Service Creation") and use the sigmac tool to convert them into **Elastic KQL**.

### Example Elastic KQL:

```
``` process.name : "powershell.exe" and process.args : ("*Invoke-Expression*" or
"*iex*" or "*-enc*") ```
```

- **Tuning:** All new detections enter a 1-week "tuning" phase in Jira to ensure they do not flood the SOC with false positives.

## 9. PLAYBOOKS & PROCESSES (JIRA INTEGRATION)

Every hunt must be documented in **Jira** for auditability and knowledge sharing.

- **The Jira Workflow:**

Draft Hunt -> Data Gathering -> Analysis -> Detection Candidate -> Completed.

- **Jira Ticket Structure:**

**Ticket Title: [HUNT] <ATT&CK Technique> – <Hunt Name>**

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| Hypothesis              | Describe the adversary behavior.                       |
| Hunt Type               | Unstructured / Situational / Data Analysis             |
| Log Sources             | List Elastic indices used (e.g.,logs-windows.sysmon-*) |
| Queries                 | Paste the KQL used during the hunt.                    |
| Findings                | True Positive / Suspicious / Benign                    |
| Feedback & tuning notes | -                                                      |

## 10. CLOSING STATEMENT

Building a **Threat Hunting** capability is not a one-time project, but a fundamental shift in how security operations are conducted. By moving from a reactive “wait-and-see” posture to a **proactive**, intelligence-led model, adversaries are forced to operate under increased scrutiny and reduced opportunity.

Through disciplined use of the **Elastic Stack** for visibility, **Jira** for structured process accountability, and **Cyber Threat Intelligence** (CTI) for prioritization and strategic direction, the SOC evolves into a mature, behavior-focused defensive function capable of addressing modern and sophisticated threats.

***The objective is clear: Increase detection precision, reduce dwell time, strengthen defensive coverage, and raise the operational cost of attack beyond its potential reward.***